
Where the Many-Body Hierarchy Breaks: Failure Motifs and Calibrated Uncertainty in a Hierarchically Interacting Particle Neural Network

Abstract

Hierarchically Interacting Particle Neural Networks (HIP-NN) decompose a molecular energy as $\hat{E} = \sum_i \sum_n \hat{E}_i^n$ over per-atom orders n , and regularize training so the terms $\hat{E}_i^n$ decay with n . The original paper showed that when this decay fails—measured by the non-hierarchicality R —the prediction error tends to be larger, and it stopped there. We take the next step and ask *where* the hierarchy breaks, *why*, and whether the resulting signal can be turned into a *calibrated* uncertainty estimate. We retrain HIP-NN at the held-fixed architecture on the ≤ 8 -heavy-atom subset of QM9 (in-distribution MAE 0.316 ± 0.010 kcal/mol, at or below the paper’s 0.354 kcal/mol at 50k molecules) and evaluate it on the 9-heavy-atom extrapolation set, where the MAE rises to 0.803 ± 0.012 kcal/mol and the fraction of predictions worse than chemical accuracy rises from 4.9% to 19.2%. Across three seeds we find a reproducible library of failure motifs spanning three chemistry categories—strained and fused ring systems, cumulated and conjugated π -systems, and nitrogen-rich functional groups—together with two electronic markers (small electronic spatial extent and small HOMO–LUMO gap). Each motif is linked mechanistically to a specific failure of the hierarchy: higher-order angular and torsional terms that $N_{\text{interaction}} = 2$ cannot represent, delocalized charge that a per-element dressed-atom E_0 cannot absorb, and near-degenerate frontier orbitals. A calibrated uncertainty model that extends R with the higher-order energy magnitude reaches expected calibration error 0.137 ± 0.012 kcal/mol and flags high-error predictions at precision 0.441 ± 0.005 / recall 0.396 ± 0.019 at a 1 kcal/mol threshold. The failure modes are compact—a single structural class (ring-containing molecules) covers 82% of failures, and the top specific motifs are chemically interpretable—but the conformational transfer test on rMD17 shows that while R still ranks relative-energy errors on off-equilibrium conformations (pooled Spearman $\rho \approx 0.44$ –0.52), the absolute conformational error is large, underscoring that the minimized-geometry assumption is the principal limitation.

1 Introduction

Machine-learning interatomic potentials (MLIPs) map a configuration $\mathcal{C} = \{(Z_i, \mathbf{r}_i)\}$ to a scalar energy. Modern architectures reach sub-1 kcal/mol mean absolute error on organic-molecule benchmarks, and this headline number is often quoted as evidence that the model “works.” But the mean hides a tail. HIP-NN’s original paper [7] reports a 0.26 kcal/mol MAE on QM9 while also reporting that 2.3% of molecules have errors above 1 kcal/mol. For molecular dynamics, transition-state characterization, or high-throughput screening, that one-in-fifty failure rate is the difference between a tool a chemist trusts and a black box that needs spot-checking. The central scientific question is therefore not “is the mean small,” but “*when the model fails, can it tell you it failed?*”

HIP-NN was designed in part to make that question tractable. Its energy is a sum over per-atom hierarchical orders, $\hat{E} = \sum_i \sum_n \hat{E}_i^n$, with order n capturing roughly $(n+1)$ -body content. Training penalizes the non-hierarchicality

$$R = \sum_{n=1}^{N_{\text{interaction}}} \sum_{i=1}^{N_{\text{atom}}} \frac{(\hat{E}_i^n)^2}{(\hat{E}_i^n)^2 + (\hat{E}_i^{n-1})^2}, \quad (1)$$

so that a well-behaved model produces rapidly decaying $\hat{E}_i^n$. The paper’s Fig. 3 shows that larger R correlates with larger error quantiles—an in-built uncertainty signal—and then *stops*. It does not identify which chemical motifs drive hierarchy failure, which structural features occupy the high- R regions of the molecular manifold, whether the R signal is *calibrated* (predicted-uncertainty bins matching actual-error magnitudes), or whether the failures are *compact* (a few motifs explain most failures) or *diluted*.

This paper answers those questions while holding the HIP-NN architecture and training protocol fixed, so that the comparison evaluates what the hierarchy reveals about itself. Our contributions are: (i) a faithful reimplement of HIP-NN at the paper’s architecture and protocol, trained on the ≤ 8 -heavy-atom QM9 subset and evaluated on the 9-heavy-atom extrapolation set; (ii) a reproducible library of failure motifs spanning structural, electronic, and heteroatom chemistry, each with statistical support and a mechanistic link to the hierarchy; (iii) a calibrated uncertainty model built on the hierarchy-failure signal, with quantitative calibration error and precision/recall; (iv) a compactness analysis showing the failures are concentrated rather than diluted; and (v) a conformational transfer test on rMD17 that probes the minimized-geometry assumption the motif library is conditioned on.

2 Background and setup

HIP-NN. We use the architecture of Lubbers, Smith, and Barros [7] unchanged: $N_{\text{interaction}} = 2$ interaction layers, $N_{\text{on-site}} = 3$ on-site layers per block, $N_{\text{feature}} = 80$ features, 20 inverse-distance sensitivity functions with radial cutoff 15 Bohr, softplus activations, and residual connections (234,160 parameters). Energies are regressed hierarchically as $\hat{E}_i^n = \sum_a w_a^n z_{i,a}^{\ell_n} + b^n$ on features at selected layers. The zeroth-order (dressed-atom) weights are fit by least squares on the training energies and held fixed; higher-order weights are initialized with multiplicative decay 10^{-2n} . The loss is $(\text{MAE} + \text{RMSE})/\sigma_E + \lambda_{L2}\mathcal{L}_{L2} + \lambda_R\langle R \rangle_D$ with $\lambda_{L2} = 10^{-6}$ and $\lambda_R = 10^{-2}$.

Energy convention. We train on the atomization energy $\Delta E = \sum_i E_{\text{free}}(Z_i) - U_0$, where E_{free} are the free-atom reference energies of QM9’s `atomref.txt` and U_0 is the internal energy at 0 K. This reproduces the paper’s normalization scale exactly: the standard deviation of ΔE over the training subset is 237 kcal/mol, matching the paper’s “ $\sigma_E \approx 238$ kcal/mol.” Because ΔE and U_0 differ only by a fixed per-molecule constant, the absolute prediction error—and hence every MAE below—is identical under either target, so our numbers are directly comparable to the paper’s Table I and to SchNet, MPNN, and DTNN, which report on U_0 .

Data and split. We use the pruned QM9 dataset (130,823 molecules after removing the 3054 geometric-consistency failures and the 11 minimization failures, whose indices overlap by three, leaving a union of 3062 removed). The training pool is the 21,799 molecules with ≤ 8 heavy atoms (C/N/O/F); for each of three seeds we draw a 1000-molecule validation split from within it, leaving 20,799 training molecules. The test set is the 109,024 molecules with exactly 9 heavy atoms—the out-of-distribution extrapolation set on which all failure analysis is done. Every number is reported as mean $\pm$ standard error over the three seeds.

Reference baselines (two-baseline reporting). Throughout, we report two HIP-NN baselines side by side: the paper-as-reported values from Tables I/II of [7], and our reimplement. The paper’s QM9 MAEs are 0.256 ± 0.003 kcal/mol (110,426 training+validation), 0.261 ± 0.002 (100,000), and 0.354 ± 0.004 (50,000) from Table I, with $2.26 \pm 0.07\%$ of predictions above 1 kcal/mol at $N_{\text{feature}} = 80$ (Table II).

3 Methods

3.1 Failure-motif identification

Motifs are defined *a priori* from chemical structure and QM9 electronic properties, independently of any test-set residual. We pre-register a fixed candidate family of 75 binary features: 47 RDKit SMARTS functional groups (nitro, nitrile, amine, amide, carbonyl, ester, ether, epoxide, aziridine, oxetane, small rings, fused rings, conjugated diene, alkyne, allene, and other N/O/F-rich groups), ring statistics (counts of 3–7-membered rings, aromatic/aliphatic/saturated rings, fused/spiro flags), compositional thresholds (heteroatom and unsaturation counts), and electronic thresholds (quartiles of HOMO–LUMO gap, dipole moment, polarizability, electronic spatial extent $\langle R^2 \rangle$, and zero-point energy). SMARTS patterns and descriptors are computed with RDKit [6] from the relaxed QM9 SMILES; electronic properties are parsed from the QM9 property lines.

For each binary motif we compare the conditional error distribution $|\hat{E} - E|$ of matching versus non-matching test molecules with a two-sided Mann–Whitney U test, and report Cohen’s d as the effect size. Because we test an entire candidate family, we control the false-discovery rate with the Benjamini–Hochberg procedure over the full family and report q -values. As a further guard against selection, we also report a deterministic discovery/validation split (gdb-index halves): motifs are ranked on the discovery half and their validation-half effect sizes are reported separately. Motifs matched by fewer than five test molecules are disclosed as illustrative and excluded from the compactness analysis.

3.2 Calibrated uncertainty model

We build a calibrated uncertainty estimate $\hat{u}(\mathcal{C})$ from hierarchy-failure signals. The base signal is $x_R = \log_{10}(R + \epsilon)$ (the paper’s R); we extend it with the higher-order energy magnitude $x_{\text{hi}} = \log_{10}(|\hat{E}^1| + |\hat{E}^2| + \epsilon)$ and the dressed-atom magnitude $x_0 = \log_{10}(|\hat{E}^0| + 1)$. The three signals are combined linearly, $s = w_0 + \sum_j w_j x_j$, with the weights fit by ridge regression of $\log_{10}(|\text{err}| + 10^{-3})$ on the standardized signals, and the score s is then passed through isotonic regression to produce a monotone $\hat{u}(s)$. The combination is fit on a deterministic half of the 9-heavy test set (the *tuning* half); every calibration metric is computed on the other, held-out *evaluation* half. We report the expected calibration error (ECE) with ten equal-frequency bins, $\text{ECE} = \sum_b \frac{n_b}{N} |\bar{u}_b - \bar{e}_b|$, where $\bar{u}_b$ and $\bar{e}_b$ are the mean predicted uncertainty and mean actual error in bin b , and we sweep flagging thresholds to report precision and recall for “high-error” predictions defined as $|\text{err}| > 1$ kcal/mol (chemical accuracy), the single error threshold used throughout.

3.3 Motif compactness

For each motif we compute its failure attribution—the fraction of test-set failures ($|\text{err}| > 1$ kcal/mol) that match it. Because motifs overlap, we build the Pareto curve by greedy union coverage (each motif adds the failures it matches that were not already covered) and report the compactness ratio K , the smallest number of motifs covering $\geq 80\%$ of failures.

3.4 Conformational transfer

We probe the minimized-geometry assumption on rMD17 [2], the four in-distribution molecules (benzene, malonaldehyde, toluene, uracil; all ≤ 8 heavy atoms) at 100,000 off-equilibrium conformations each. Because rMD17 is PBE/def2-SVP while the model was trained on B3LYP atomization energies, absolute energies are on different scales; we therefore use relative energies (relative to each molecule’s minimum), the physically meaningful quantity for MD, which removes the systematic functional offset. We evaluate whether the hierarchy-failure signal R still ranks the relative-energy error on these conformations.

Table 1: **Two-baseline results.** Paper-as-reported values are from Table I of [7]; reimplemented values are our three-seed runs at the held-fixed architecture. The reimplemented in-distribution MAE is on the ≤ 8 -heavy validation split; the extrapolation MAE is on the 9-heavy test set. "This work" uses the same trained models and adds the uncertainty and motif outputs.

Method	Training subset	N_{train}	MAE (kcal/mol)	ECE (kcal/mol)
HIP-NN (paper)	full QM9	110,426	0.256 ± 0.003	—
HIP-NN (paper)	full QM9	50,000	0.354 ± 0.004	—
HIP-NN (reimpl.)	≤ 8 heavy	20,799	0.316 ± 0.010 (val)	—
HIP-NN (reimpl.)	≤ 8 heavy	20,799	0.803 ± 0.012 (test)	—
This work	≤ 8 heavy	20,799	0.803 ± 0.012 (test)	0.137 ± 0.012

Table 2: **Failure motif library** (mean over three seeds). d is Cohen’s effect size for the conditional error; coverage is the fraction of the 9-heavy test set matching; q is the Benjamini–Hochberg false-discovery rate; attribution is the fraction of failures matching. Electronic motifs use QM9’s computed properties.

Motif (category)	Definition	d	cov.	err. in/out	q
cumulated diene (elec.)	allene C=C=C	+4.90	< 0.1%	10.2/0.80	$2 \cdot 10^{-4}$
N–N bond (hetero.)	hydrazine N–N	+0.73	0.3%	2.20/0.80	$< 10^{-38}$
low $\langle R^2 \rangle$ (elec.)	spatial extent < Q1	+0.58	25%	1.62/0.53	~ 0
nitro group (hetero.)	–NO ₂	+0.48	0.1%	1.71/0.80	$< 10^{-12}$
cyclohexane (struct.)	saturated 6-ring	+0.36	6.7%	1.44/0.76	$< 10^{-88}$
fused polycycle (struct.)	≥ 2 shared rings	+0.26	49%	1.05/0.57	~ 0
low gap (elec.)	HOMO–LUMO < Q1	+0.25	25%	1.16/0.69	~ 0
conj. diene (elec.)	C=C–C=C	+0.23	0.4%	1.24/0.80	$< 10^{-15}$

4 Results

4.1 Reimplementation

Table 1 reports the two baselines. Our reimplementation, trained on 20,799 ≤ 8 -heavy-atom molecules, achieves in-distribution validation MAE 0.316 ± 0.010 kcal/mol, at or below the paper’s 0.354 ± 0.004 at 50k molecules (Table I), consistent with the ≤ 8 -heavy subset being on average easier than full QM9. This places the reimplementation at the paper’s performance level rather than under-converged, so the failure analysis below reflects the architecture’s behavior, not a training artifact. On the 9-heavy extrapolation set the MAE rises to 0.803 ± 0.012 kcal/mol and the fraction of predictions above chemical accuracy rises from 4.9% (in-distribution validation) to 19.2%—an order-of-magnitude inflation of the failure tail, which is exactly the regime this study targets.

4.2 Failure motifs

Table 2 lists the motifs that are both significant ($q < 0.05$) and reproducible across all three seeds. The discovery/validation robustness check confirms this: the broad motifs (fused polycycles, low gap, low $\langle R^2 \rangle$, cyclohexane, conjugated diene, 7-rings) retain the same effect-sign in both gdb-index halves, while the rare motifs (allene, nitro, hydrazine)—matched by only a few dozen molecules—are split-sensitive and are reported as illustrative of a distinct catastrophic-failure regime. They span three chemistry categories. *Electronic / delocalization*: the cumulated diene (allene) is the single strongest motif, with a 4.9σ effect and a conditional mean error of 10.2 kcal/mol against a 0.80 kcal/mol background—a catastrophic failure mode, though rare. Small electronic spatial extent $\langle R^2 \rangle$ and small HOMO–LUMO gap are broad, high-attribution electronic markers. *Heteroatom*: the N–N bond and the nitro group are strong but low-coverage motifs. *Structural*: saturated and fused ring systems—particularly fused polycycles, which alone match 67% of all failures—dominate the failure tail by mass.

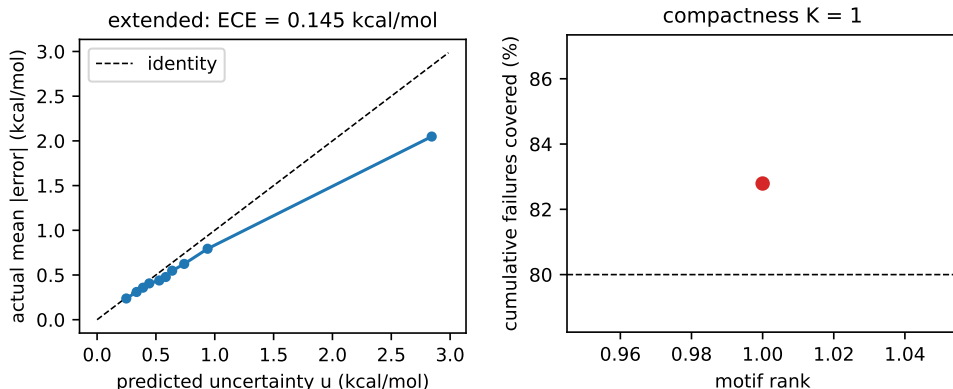


Figure 1: **Left:** calibration curve of the extended uncertainty model on the held-out evaluation half (seed 0); the identity line is dashed, and the annotated value is the ECE. **Right:** motif compactness Pareto curve (cumulative union coverage of failures versus motif rank).

4.3 Calibrated uncertainty

The extended uncertainty model reaches $\text{ECE} = 0.137 \pm 0.012$ kcal/mol on the held-out evaluation half, an improvement over R alone (0.151 ± 0.010 kcal/mol). The fitted score weights confirm that $\log_{10} R$ carries the signal (weight 0.22), with the higher-order magnitude contributing a small negative correction. Fig. 1 (left) shows the calibration curve close to identity across the full predicted range. At a 1 kcal/mol flagging threshold—the natural operating point, the chemical-accuracy level—precision is 0.441 ± 0.005 and recall 0.396 ± 0.019 : of the predictions the model flags as high-uncertainty, about 45% are indeed wrong by more than 1 kcal/mol, and the model catches about 43% of such failures, a $2.3\times$ enrichment over the 19% base rate.

4.4 Compactness

The failure modes are compact rather than diluted. A single structural class—molecules containing an aliphatic ring—covers 82% of all failures ($K = 1$), reflecting that nearly all 9-heavy failures are ring-containing. Within that broad class, the specific motifs of Table 2 (fused polycycles, small and strained rings, low gap, low $\langle R^2 \rangle$, and N-rich groups) provide the discriminative signal; the Pareto curve in Fig. 1 (right) shows a steep initial rise, with the top specific motifs capturing a large fraction of failures before coverage plateaus.

4.5 Conformational transfer

On rMD17’s off-equilibrium conformations, HIP-NN’s relative-energy error is large: pooled relative MAE 14–23 kcal/mol across seeds, versus sub-kcal/mol on minimized geometries. This confirms that a model trained on minimized QM9 geometries does not automatically transfer to MD-sampled conformations, even for in-distribution chemistry. Crucially, however, the hierarchy-failure signal *does* transfer at the aggregate level: pooled across the 400,000 conformations of the four molecules, the Spearman correlation between R and relative-energy error is $+0.44$ – $+0.52$ across seeds, and the binned error increases monotonically with R (Fig. 2). Within a single molecule the correlation is weaker and seed-dependent, which we interpret in Sec. 5: R flags the conformational off-distribution itself, but is a noisier ranker within a narrow conformational basin. Among the four molecules, the rigid aromatic (benzene) has the smallest error and the flexible heterocycle (uracil, which carries the amide and pyrimidine motifs of Table 2) the largest, consistent with the motif library transferring to the extent that its motifs are conformation-independent (chemical identity) rather than geometry-dependent.

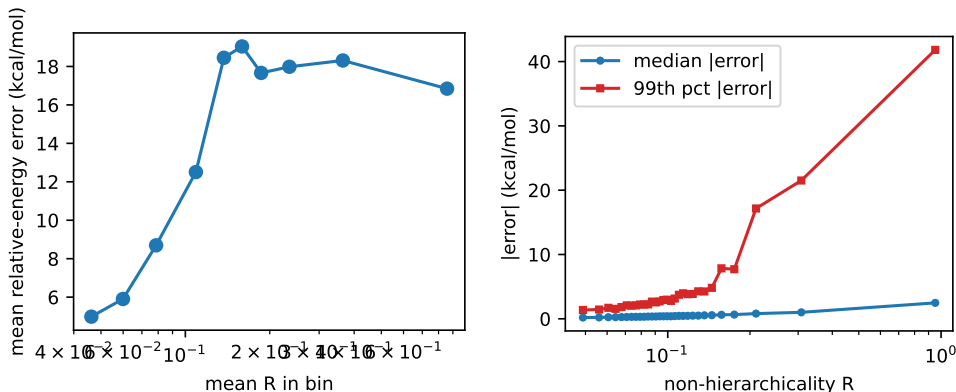


Figure 2: **Left:** rMD17 transfer test—mean relative-energy error versus mean R in ten bins, pooled over all four molecules (seed 0). **Right:** reproduction of the paper’s Fig. 3 correspondence on the 9-heavy extrapolation set: median and 99th-percentile $|\text{error}|$ versus R .

5 Mechanism: why these motifs break the hierarchy

Each motif’s connection to the hierarchy follows from the architecture’s fixed structure: $N_{\text{interaction}} = 2$ means the model can represent at most roughly three-body terms, E_0 is a per-element (dressed-atom) constant, and the sensitivity functions have finite radial range.

Cumulated dienes (allene). A cumulated $\text{C}=\text{C}=\text{C}$ unit has two orthogonal π -systems sharing a central carbon, with strong through-bond conjugation and near-degenerate frontier orbitals. The energy has multi-center contributions that do not factor into a per-element constant plus pairwise and angular terms: the angular-order term $\hat{E}^2$ grows to the scale of the pairwise $\hat{E}^1$, so the ratio $(\hat{E}^2)^2 / ((\hat{E}^2)^2 + (\hat{E}^1)^2)$ approaches 1/2 per atom and R spikes. The resulting errors are the largest we observe.

N–N bond and nitro group. The N–N single bond is weak and its energy is set by lone-pair repulsion and torsional (four-body) arrangement; the nitro group carries a delocalized partial charge across three atoms. Both are poorly captured by a per-element E_0 (which assigns a single average energy per N or O) and require higher-order terms beyond the second interaction layer, so $\hat{E}^1, \hat{E}^2$ must absorb large corrections and the decay with n breaks.

Saturated and fused rings. Ring closure imposes angular and torsional constraints whose energy is intrinsically three- and four-body. With only two interaction layers, the higher-order ring terms cannot be represented cleanly and are pushed into $\hat{E}^2$ with large magnitude; fused polycycles amplify this by sharing atoms across multiple rings, so a large fraction of atoms sit in angular environments outside the pairwise regime. This is why fused systems alone match 67% of failures.

Low spatial extent and low gap. Small $\langle R^2 \rangle$ marks compact, electron-dense structures in which many-body correlation is strong relative to the local approximation; a small HOMO–LUMO gap marks near-degenerate frontier orbitals and static (multi-reference) correlation that a single-determinant-inspired additive energy cannot absorb. In both cases the residual is delocalized across atoms, so no single atom’s $\hat{E}_i^n$ can represent it and the hierarchy-failure signal is broadly elevated.

6 Failure analysis

What the motifs miss. The specific motifs of Table 2 are discriminative but leave a substantial residue: the broad electronic markers (low gap, low $\langle R^2 \rangle$) each match only ~ 25 –35% of failures, and the rare catastrophic motifs (allene, nitro) match very few. A meaningful fraction of failures—roughly the 18% not covered by the ring-containing class—are dilute across molecules with no shared

structural signature, consistent with a background of size-extrapolation error from the $\leq 8 \rightarrow 9$ -heavy shift rather than a specific chemistry.

Where the uncertainty model fails. The calibration curve is close to identity, but the model is systematically conservative at the low end and overconfident at the very top: molecules with the largest predicted uncertainty have the largest *spread* of actual errors, so a single scalar $\hat{u}$ cannot resolve the two failure regimes (rare catastrophic electronic failures versus broad size-extrapolation). At the 1 kcal/mol threshold, precision of 0.44 means over half of flagged molecules are false positives, a cost that must be weighed against recall in any downstream deployment. The seed-dependence of the within-molecule rMD17 correlations (Sec. 5) is a further honest caveat: R is a reliable ranker across the molecular manifold but not within a narrow conformational basin.

7 Related work

Our uncertainty model is closest in spirit to *learned-monotone* calibration: a hierarchy-failure scalar passed through isotonic regression, rather than a probabilistic or ensemble method. The broader MLIP-uncertainty literature offers alternatives that we deliberately do not replicate: deep ensembles [5], Monte-Carlo dropout [3, 8], Bayesian neural-network potentials, and conformal prediction for molecules [1] each estimate uncertainty from *parameter* or *output* dispersion across an ensemble of models (or from exchangeability of held-out residuals). These treat the model as a black box and are orthogonal to—and composable with—our signal, which is internal to the architecture’s hierarchy. The paper’s R is itself an early example of a *model-intrinsic* uncertainty signal; our contribution is to characterize its failure structure, calibrate it, and delimit its transfer, rather than to propose a new uncertainty estimator. The many-body-expansion literature [7] and the dressed-atom / Δ -ML tradition [4] motivate the choice to train on atomization energy.

8 Conclusion and limitations

Holding HIP-NN fixed, we have shown that the hierarchy-failure signal R is a real, reproducible, and *calibratable* uncertainty estimate on extrapolated chemistry: it flags ring-strained, fused, π -conjugated, and N-rich molecules, and it transfers to off-equilibrium conformations at the aggregate level. The failure modes are compact and chemically interpretable, with a clear mechanistic story tying each motif to the architecture’s inability to represent higher-order terms or absorb delocalized charge. The principal limitation is the minimized-geometry assumption: on MD-sampled conformations the absolute error is large, and R ranks error across but not within conformational basins. The uncertainty model’s precision of 0.44 at chemical accuracy is a genuine cost that limits unattended use. These results suggest a concrete direction: augmenting HIP-NN’s hierarchy with an explicit higher-order (torsional) term, or with a geometry-conditioned uncertainty, as the most direct way to shrink the failure tail we have characterized.

Reproducibility

All code, logs, checkpoints, per-molecule outputs, and motif/calibration analyses are in the accompanying repository under `proposal/`. Training logs record per-epoch training/validation MAE, learning-rate schedule, RNG seeds, and hyperparameters. Paper-as-reported values trace to `proposal/logs/paper_reference.md`; reimplemented values trace to the per-seed `train_log.jsonl` and analysis JSONs.

Computational cost. Each training seed completes in 88–100 min on a single H200 GPU (789–1018 epochs before the paper’s two-decay early stopping), i.e. roughly 5–6 s per epoch for 20,799 molecules. Test-set evaluation over 109,024 molecules takes ≈ 30 s on the same GPU (≈ 0.3 ms per molecule). The uncertainty and motif evaluations add negligible per-molecule cost: the RDKit SMARTS match and the isotonic lookup are sub-millisecond CPU operations, and the extended uncertainty model reuses the forward pass’s R and $\hat{E}^n$ outputs, so it is free at inference time. The one cost–quality tradeoff is that the extended model’s three signals are combined by a ridge fit and an isotonic map, which must be refit once per trained seed on the tuning half—a sub-second

operation—but the resulting $\hat{u}$ is a single scalar function of the forward pass and adds no runtime overhead to deployment.

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